

THE BREUIL-KISIN TWIST

CHARLIE WU

ABSTRACT. We discuss the Breuil-Kisin twist.

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1. WHAT DID WE DO LAST TIME?

Last time, Jason described the prismatic site attached to a p -adic formal scheme X . Roughly, X_Δ consists of prisms (A, I) where A is a δ -ring which is (p, I) -adically complete, I is a principal divisor, and $\varphi(x) := x^p + p\delta(x)$ is a lift of Frobenius of A/I . There a structure sheaf $\mathcal{O}_\Delta$ given by sending a prism (A, I) to A . We can therefore discuss the notion of a vector bundle. Let $\text{Vect}(X_\Delta, \mathcal{O}_\Delta)$ be the category of vector bundles on $(X_\Delta, \mathcal{O}_\Delta)$.

2. PRISMATIC F -CRYSTALS

We want to study local systems on $\text{Spec } k$ by instead passing through some sort of Riemann-Hilbert type correspondence. The output of this correspondence takes a local system and produces a coherent object (some kind of vector bundle). We are going to describe a special kind of coherent object which will be important to us later.

Definition 2.1. Let X be a p -adic formal scheme with a lift of Frobenius φ . Write $\text{Vect}^\varphi(X_\Delta, \mathcal{O}_\Delta)$ to be the category of *prismatic F -crystals* on X_Δ . The objects are given by (E, φ_E) where E is a vector bundle on $(X_\Delta, \mathcal{O}_\Delta)$, and $\varphi_E : \varphi^*E[1/I_\Delta] \rightarrow E[1/I_\Delta]$ is an isomorphism.

There is only one notion of morphism which makes sense for this category. A morphism $f : (E, \varphi_E) \rightarrow (F, \varphi_F)$ is a morphism $f : E \rightarrow F$ such that the diagram

$$\begin{array}{ccc}
 \varphi^*E[1/I_\Delta] & \xrightarrow{\varphi^*f} & \varphi^*F[1/I_\Delta] \\
 \varphi_E \downarrow & & \downarrow \varphi_F \\
 E[1/I_\Delta] & \xrightarrow{f} & F[1/I_\Delta]
 \end{array}$$

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commutes. Giving $\text{Vect}^\varphi(X_\Delta, \mathcal{O}_\Delta)$ a tensor product makes it into a symmetric monoidal category.

Which local systems do these prismatic F -crystals correspond to? In the case where $X = \text{Spf}(\mathcal{O}_K)$, we have the following theorem which tells us:

Theorem 2.2 (Bhatt-Scholze). *There is an equivalence of categories*

$$\text{Vect}^\varphi(\text{Spf}(\mathcal{O}_K)_\Delta, \mathcal{O}_\Delta) \cong \{\text{crystalline } \mathbb{Z}_p\text{-local systems on } \text{Spec } K\}$$

via an “étale realization functor” (to be discussed in a future talk).

There is another way to understand prismatic F -crystals open-by-open.

Definition 2.3. Let X be a p -adic formal scheme. There is a natural equivalence

$$\lim_{(A,I) \in X_\Delta} \text{Vect}(A) \cong \text{Vect}(X_\Delta, \mathcal{O}_\Delta).$$

Then, a prismatic F -crystal is a collection of A -modules E_A for every $(A, I) \in X_\Delta$ with identifications $E_A \otimes_A B \cong E_B$ whenever $(B, J) \rightarrow (A, I)$ is a morphism, and for every $(A, I) \in X_\Delta$ an identification $\varphi_A : \varphi^* E_A[1/I] \rightarrow E_A[1/I]$.

3. BREUIL-KISIN TWISTS

If we want a category of coherent objects corresponding to some class of local systems, we need to understand what the Tate twist is in that category. The correct Tate twist is called the Breuil-Kisin twist. To define this twist in general, we define it for $\text{Spf}(\mathbb{Z}_p)$ first and then pull back.

Definition 3.1 (psuedo-definition). Let $X = \text{Spf}(\mathbb{Z}_p)$. There is a natural prismatic F -crystal $(\mathcal{O}_\Delta\{1\}, \varphi)$ where

$$\mathcal{O}_\Delta\{1\} := I_\Delta \otimes \varphi^* I_\Delta \otimes (\varphi^2)^* I_\Delta \otimes \cdots = \bigotimes_{n \geq 0} (\varphi^n)^* I_\Delta.$$

Then, formally we have the property that

$$\mathcal{O}_\Delta\{1\} \otimes I_\Delta^{-1} \cong \varphi^* \mathcal{O}_\Delta\{1\}$$

which shows that $\mathcal{O}_\Delta\{1\}$ is a prismatic F -crystal.

Remark 3.2. The above (infinite) tensor product does not make sense, so we need to fix it.

3.3. Transversal prisms.

Definition 3.4. A prism (A, I) is *transversal* if A/I is p -torsion free. Equivalently, A is p -torsion free. Equivalently, the natural map $I/pI \rightarrow A/pA$ is injective.

Example 3.5 (The q -de Rham Prism). Let $A = \mathbb{Z}_p[[q-1]]$ be the completion of $\mathbb{Z}[q]$ with respect to the ideal $(p, q-1)$. We give A the structure of a δ -ring with Frobenius lift

$$\varphi(q) \mapsto q^p.$$

Then, $(A, I) = (A, ([p]_q))$ where $[p]_q$ is the element

$$[p]_q = \frac{q^p - 1}{q - 1} = 1 + q + \cdots + q^{p-1}.$$

Then,

$$A/I = \mathbb{Z}_p[[q-1]]/([p]_q) \cong \mathbb{Z}_p[q]/(1 + q + \cdots + q^{p-1}).$$

This is clearly a p -torsion free ring; for example, it is isomorphic to the ring of integers in $\mathbb{Q}_p(e^{2\pi i/p})$.

Therefore, (A, I) is a transversal prism.

Lemma 3.6. *Let (A, I) be a transversal prism. For every $r \geq 0$, the inclusion $I \hookrightarrow A$ induces a monomorphism $\theta : (\varphi_A^r)^*(I) \rightarrow (\varphi_A^r)^*(A) \cong A$. And, $\text{coker } \theta$ is p -torsion free.*

Proof. ♠♠♠ CHARLIE: [Maybe omit this?] Since $(\varphi_A^r)^*(I)$ is an invertible A -module, it is p -complete and p -torsion free. Therefore, it is enough to show that the induced map $(\varphi_A^r)^*(I/pI) \rightarrow A/pA$ is injective since we can just check limits.

But this is clear, since I/pI is an invertible ideal in A/pA . Therefore, the Frobenius pullback maps isomorphically to $(\varphi_A^r)^*(I)$. $\square$

Let (A, I) be a transversal prism, and let $r \geq 0$. Then, the Frobenius lift $\varphi_A : A \rightarrow A$ allows us to view $(\varphi_A^r)^*(I)$ as an ideal in A .

We let $I_r = I \otimes_A \varphi_A^*(I) \otimes_A \cdots \otimes_A (\varphi_A^{r-1})^*(I) \subseteq A$.

Then,

Lemma 3.7. *Let (A, I) be a transversal prism. Then A is an inverse limit of quotient rings*

$$\cdots \twoheadrightarrow A/I_3 \twoheadrightarrow A/I_2 \twoheadrightarrow A/I_1.$$

and A is also an inverse limit of quotient rings

$$\cdots \twoheadrightarrow A/\varphi_A^*(I_3) \twoheadrightarrow A/\varphi_A^*(I_2) \twoheadrightarrow A/\varphi_A^*(I_1).$$

Lemma 3.8. *Let (A, I) be a transversal prism, and let $r \geq 0$. Then,*

- (1) *The A -module I_r/I_r^2 is p -torsion free.*
- (2) *The canonical map $I_{r+1}/I_{r+1}^2 \rightarrow I_r/I_r^2$ is divisible by p .*
- (3) *The induced map $I_{r+1}/I_{r+1}^2 \xrightarrow{1/p} I_r/I_r^2$ is surjective.*

Definition 3.9. Let (A, I) be a transversal prism. We define the Breuil-Kisin twist $A\{1\}$ to be

$$A\{1\} = \varprojlim (\cdots \xrightarrow{1/p} I_3/I_3^2 \xrightarrow{1/p} I_2/I_2^2 \xrightarrow{1/p} I_1/I_1^2).$$

This construction is functorial. If $f : (A, I) \rightarrow (B, J)$ is a morphism of transversal prisms, then there is a unique map $f\{1\} : A\{1\} \rightarrow B\{1\}$ so that for all $r \geq 0$, the diagram

$$\begin{array}{ccc} A\{1\} & \xrightarrow{f\{1\}} & B\{1\} \\ \downarrow & & \downarrow \\ I_r/I_r^2 & \xrightarrow{f} & J_r/J_r^2 \end{array}$$

is commutative. Furthermore, $f\{1\} : A\{1\} \otimes_A B \rightarrow B\{1\}$ is an isomorphism. The Frobenius on $A\{1\}$ is defined as follows. We observe that

$$\begin{aligned} \varphi_A^*(I_r) &= \varphi_A^*(I \otimes_A \varphi_A^*(I) \otimes_A \cdots \otimes_A (\varphi_A^{r-1})^*(I)) \\ &= I^{-1} \otimes_A I \otimes_A \varphi_A^*(I) \otimes_A (\varphi_A^2)^*(I) \otimes_A \cdots \otimes_A (\varphi_A^r)^*(I) \\ &\cong I \otimes_A I_{r+1}. \end{aligned}$$

Then, we observe that $\varphi_A^*(A\{1\}) \otimes_A A/\varphi_A^*(I_r) \cong I^{-1} \otimes_A A\{1\} \otimes_A A/\varphi_A^*(I_r)$. But since passing to limits yields $\varprojlim_r A/(\varphi_A)^*(I_r) \cong A$, we obtain an isomorphism $\varphi_A^*(A\{1\}) \cong I^{-1} \otimes_A A\{1\}$. This, we call the Frobenius.

3.10. Extension to non-transverse prisms. To extend to non-transverse prisms, we bootstrap from the transverse case.

Lemma 3.11. *Let (A, I) be a prism. Then, there exists a prism homomorphism $(B, J) \rightarrow (A, I)$ where (B, J) is transversal.*

Proposition 3.12 (Proposition-definition). *Let $\mathcal{C}$ be the category whose objects are transversal prisms (B, J) with a morphism of prisms $f : (B, J) \rightarrow (A, I)$. Then, the diagram*

$$(B, J) \in \mathcal{C} \mapsto A \otimes_B B\{1\}$$

has a direct limit in the category of A -modules. This resulting direct limit we call $A\{1\}$.

4. PRISMATIC LOGARITHM

The Tate twist appears naturally in étale cohomology in the following way. Say ℓ is invertible in k . Then, there is a chern class map

$$c_1 : H_{\text{ét}}^1(X, \mathbb{G}_m) \cong \text{Pic}(X) \rightarrow H_{\text{ét}}^2(X, \mathbb{Z}/\ell(1))$$

coming from the Kummer exact sequence. This is some kind of analogue to the exponential exact sequence. The connecting homomorphism is a kind of logarithm. In the complex setting, it is given by $d \log$. We seek an a similar kind of construction via a map $T_p(\mathbb{G}_m(A/I)) \rightarrow A\{1\}$.

Let $u \in T_p(\mathbb{G}_m(A/I))$. This is a sequence of elements $u_0, u_1, \dots$ such that $u_0 = 1$ and $u_n = u_{n+1}^p$. We can uniquely lift this to a sequence of elements $v_0, v_1, \dots$, so that $v_i = u_i \pmod{I}$.

Here, v_0 is not always 1 but it satisfies the condition that $\delta_A(v_0) = 0$. This is because $\varphi_A(v_0) = v_0^p + p\delta(v_0) = 1$. The elements v satisfying $\delta_A(v) = 0$ are called rank 1 elements.

Let $(1 + I)_{\text{rk}=1}$ be the collection of rank 1 elements $v \in A$ such that $v = 1 \pmod{I}$, as a subgroup of $\mathbb{G}_m(A)$. Then, we get a group homomorphism

$$\rho : T_p(\mathbb{G}_m(A/I)) \rightarrow (1 + I)_{\text{rk}=1}, \quad (u_0, u_1, \dots) \rightarrow v_0.$$

We construct another map $\log_{\Delta} : (1 + I)_{\text{rk}=1} \rightarrow A\{1\}$. The composition (by abuse of notation) is also called $\log_{\Delta}$ and is called the prismatic logarithm.

4.1. The logarithm for transversal prisms. It is enough to construct the map $(1 + I)_{\text{rk}=1} \rightarrow A\{1\}$. This is done in Bhatt-Lurie, and the motivation for their construction is that if x is a real number, $\log(x) = \lim_{\alpha \rightarrow 0} (x^\alpha - 1)/\alpha$.

Proposition 4.2. *Let (A, I) be a transversal prism, and let $u \in 1 + I$ be a unit so that $\delta(u) = 0$. Then,*

- (1) *For all $r \geq 0$, $u^{p^r} = 1 \pmod{I_{r+1}}$.*
- (2) *For $r > 0$, the map*

$$\frac{1}{p} : I_{r+1}/I_{r+1}^2 \twoheadrightarrow I_r/I_r^2$$

carries the class of $u^{p^r} - 1$ to $u^{p^{r-1}} - 1$.

Proposition 4.3 (proposition-definition). *Let (A, I) be a transversal prism, and let $u \in 1 + I$ satisfy $\delta(u) = 0$. Then, by the previous proposition there is a unique element in $A\{1\}$ having image $u^{p^r} - 1$ via the quotient maps $A\{1\} \rightarrow I_{r+1}/I_{r+1}^2$.*

Call this element $\log_{\Delta}(u)$. The map $\log_{\Delta} : (1 + I)_{\text{rk}=1} \rightarrow A\{1\}$ is a group homomorphism (with group structure on the domain given by multiplication).

Proof. We need to show that $\log_{\Delta}(uv) = \log_{\Delta}(u) + \log_{\Delta}(v)$. So for each $r \geq 0$, we need to show that

$$(uv)^{p^r} - 1 = (u^{p^r} - 1) + (v^{p^r} - 1) \pmod{I_{r+1}^2}.$$

But we have that

$$(uv)^{p^r} - 1 - (u^{p^r} - 1) - (v^{p^r} - 1) = (u^{p^r} - 1)(v^{p^r} - 1)$$

and both $(u^{p^r} - 1)$ and $(v^{p^r} - 1)$ belong to I_{r+1} . $\square$

The construction is functorial in the sense that if $f : (B, J) \rightarrow (A, I)$ is a morphism of transversal prisms, then the diagram

$$\begin{array}{ccc} (1+J)_{\text{rk}=1} & \xrightarrow{f} & (1+I)_{\text{rk}=1} \\ \log_{\Delta} \downarrow & & \downarrow \log_{\Delta} \\ B\{1\} & \xrightarrow{f\{1\}} & A\{1\} \end{array}$$

commutes.

4.4. The non-transversal case. Let (A, I) be a (not necessarily transversal!) prism. Then, there is a unique function

$$\log_{\Delta} : (1+I)_{\text{rk}=1} \rightarrow A\{1\}$$

so that for every morphism of prisms $f : (B, J) \rightarrow (A, I)$ with (B, J) transversal, the diagram

$$\begin{array}{ccc} (1+J)_{\text{rk}=1} & \xrightarrow{f} & (1+I)_{\text{rk}=1} \\ \log_{\Delta} \downarrow & & \downarrow \log_{\Delta} \\ B\{1\} & \xrightarrow{f\{1\}} & A\{1\} \end{array}$$

commutes. We call this map the *prismatic logarithm*.

Lemma 4.5. *The Frobenius $\varphi_{A\{1\}} : A\{1\} \rightarrow A\{1\}$ sends $\log_{\Delta}(u)$ to $\log_{\Delta}(u)$.*

Proof. We reduce to the transversal case and note that $\varphi_A(u^{p^r} - 1) = u^{p^{r+1}} - 1$. $\square$

5. THE EXAMPLE OF THE q -DE RHAM PRISM $\mathbb{Z}_p[[q-1]]\{1\}$

We consider the concrete example of the q -de Rham prism.

Let $(A, I) = (\mathbb{Z}_p[[q-1]], ([p]_q))$. Then, observe that $q^p - 1 = (q-1)(1 + \dots + q^{p-1})$ so $q^p \equiv 1 \pmod{I}$. That is, q^p is a rank 1 element of $(1+I)^{\times}$. Then, we have an element $\log_{\Delta}(q^p) \in A\{1\}$.

Proposition 5.1. *Let (A, I) be the q -de Rham prism. There is a unique element $e_A \in A\{1\}$ satisfying $(q-1)e_A = \log_{\Delta}(q^p)$. And, $A\{1\}$ is the free A -module generated by e_A .*

Proof. The ideal I is generated by $[p]_q$. Then, I_r is generated by the products the Frobenius powers of $[p]_q$. That is,

$$I_r = I \otimes_A \varphi_A^*(I) \otimes \dots \otimes (\varphi_A^{r-1})^*(I).$$

But I is generated by $[p]_q = (q^p - 1)/(q - 1)$. Therefore, $(\varphi_A^j)^*(I)$ is generated by $(q^{p^{j-1}} - 1)/(q^{p^{j-2}} - 1)$.

Therefore, I_r is generated by

$$\left(\frac{q^p - 1}{q - 1}\right) \cdot \left(\frac{q^{p^2} - 1}{q^p - 1}\right) \cdots \left(\frac{q^{p^r} - 1}{q^{p^{r-1}} - 1}\right) = \frac{q^{p^r} - 1}{q - 1} = [p^r]_q.$$

In other words, we know that $q^{p^r} - 1 = 0 \pmod{I_r}$. Therefore, there is a unique element $e_A \in A\{1\}$ whose image in each quotient I_r/I_r^2 is the class of $[p^r]_q$. Therefore, e_A generates $A\{1\}$ since the residue classes generate I_r .

Furthermore, by definition $(q - 1)[p^r]_q = q^{p^r} - 1$ which implies that $(q - 1)e_A = \log_\Delta(q^{p^r})$. But $q - 1$ is invertible in A , so e_A is uniquely determined by this property. $\square$

In other words, $e_A = \log_\Delta(q^p)/(q - 1)$. But then, the Frobenius sends e_A to $\log_\Delta(q^p)/(q^p - 1) = [p]_q^{-1} \cdot e_A$.

Proposition 5.2. *Let $f : (A, I) \rightarrow (B, J)$ be a morphism of prisms. Let $e_B = f\{1\}(e_A)$. Then, e_B generates $B\{1\}$.*

Example 5.3. Let (B, J) be a Crystalline prism (i.e., $J = (p)$). There is a canonical map of prisms $f : (A, I) \rightarrow (B, J)$ given by $q \mapsto 1$ such that $f([p]_q) = p$.

Then, we have a canonical generator e_A with Frobenius acting by $\varphi_{A\{1\}}(e_A) = e_A/p$.

REFERENCES